MS-GPT: Rethinking MS/MS De Novo Structure Elucidation as Spectrum-Induced Posterior Querying of a Molecule-Language Model

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Abstract

Molecular structure elucidation from tandem mass spectra (MS/MS) is a central inverse problem in analytical chemistry. Most existing approaches to MS/MS identification remain tied to reference libraries or predefined candidate sets, whereas *de novo* methods aim to generate structures directly from spectra. A common *de novo* route predicts a molecular fingerprint from the spectrum and then decodes structures from it, enabling decoder pretraining on large molecule-only corpora. However, this paradigm creates a training–inference mismatch: the decoder is trained on oracle fingerprints computed from molecules, but at inference it is queried with a noisy spectrum-induced fingerprint posterior that is typically collapsed to a single thresholded fingerprint. We introduce MS-GPT, which recasts fingerprint-mediated *de novo* elucidation as spectrum-induced posterior querying of a conditional molecule-language model. MS-GPT conditions a molecule-language model on fingerprints and formulas, then converts the spectrum-induced posterior into a band of fingerprint queries near the oracle-fingerprint manifold through active-bit density calibration. Candidates sampled across this band are pooled and ranked by generation-frequency consensus. A lightweight LoRA adapter further mitigates domain-specific posterior bias while preserving the pretrained molecular prior. On NPLIB1 and MassSpecGym, MS-GPT sets a new state of the art, reaching Top-1/Top-10 exact-match accuracy of 29.8%/41.1% and 23.9%/28.7%, respectively. Candidate-pool scaling shows that efficient autoregressive molecular generation continues to improve recall with a little additional inference cost. The source code and model checkpoints are available at https://github.com/VIKI623/MS-GPT.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Chemistry*.

Keywords

tandem mass spectrometry, de novo structure elucidation, spectrum-induced posterior, molecule-language model, AI for science

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1 Introduction

Molecular structure elucidation from tandem mass spectra (MS/MS) is a central inverse problem in analytical chemistry, with applications in drug discovery, natural product research, environmental monitoring, and clinical diagnostics [1]. Existing computational approaches match query spectra to reference libraries [16], score database candidates through simulated fragmentation [34, 40], infer molecular fingerprints for candidate ranking [10, 13], or retrieve structures through cross-modal alignment [8, 18]. Although effective, these methods remain tied to reference spectra or predefined candidate sets, limiting their reach to previously characterized regions of chemical space. *De novo* structure elucidation instead generates candidate structures directly from a query spectrum, offering a route into the dark chemical space that reference-dependent methods leave unexplored [4, 6, 36].

A practical route to *de novo* structure elucidation is the two-stage fingerprint-mediated paradigm. A spectrum encoder first predicts a molecular fingerprint from the MS/MS input, and a conditional decoder then generates structures from the predicted fingerprint [36]. Because fingerprints can be computed from molecular structures alone, the decoder can be pretrained on large molecule-only corpora without annotated spectra [21, 38]. Recent work in this paradigm has largely emphasized more expressive decoders, including sequence, diffusion-like, and flow-based architectures [2, 3, 14, 28, 37].

However, improving the decoder architecture alone leaves a central mismatch unresolved: the decoder is trained on discrete oracle fingerprints computed from molecules, but inference supplies a noisy spectrum-induced fingerprint posterior π. Most pipelines collapse this posterior to a single thresholded fingerprint before molecule generation, replacing a distribution over fingerprint bits with one deterministic point query. We instead treat the posterior π not as a fingerprint to be thresholded once, but as a distribution from which decoder queries should be constructed. This view reformulates fingerprint-mediated MS/MS *de novo* elucidation as spectrum-induced posterior querying of a conditional molecule-language model: the model learns a spectrum-free map from fingerprint-formula queries to compatible molecular structures, while spectral evidence enters at inference through posterior-derived queries rather than a single point estimate.

Here, we introduce MS-GPT to realize this posterior-querying formulation. MS-GPT builds on SAFE-GPT [29], a pretrained autoregressive generator for molecular SAFE token sequences, and adapts it to condition on fingerprints and formulas. MS-GPT addresses this oracle-to-posterior gap along two complementary axes: how the posterior is converted into queries the model can answer, and how the model is adapted to answer domain-shifted posterior queries accurately. Along the first axis, active-bit density band calibration and fixed-size group querying keep posterior-derived queries near the oracle-fingerprint manifold while spanning plausible posterior operating points. Along the second axis, posterior-aligned adaptation selectively updates the query-reading pathway, aligning the model with posterior-derived active-bit co-occurrence patterns while preserving the pretrained molecular prior.

Our contributions are summarized as follows:

- We reformulate fingerprint-mediated MS/MS *de novo* elucidation as *spectrum-induced posterior querying of a conditional molecule-language model*, moving beyond the common point-query approximation. The conditional molecule-language model is pretrained on molecule-only data and conditioned on posterior-derived queries at inference.
- We propose three components for spectrum-induced posterior querying. *Active-bit density band calibration* calibrates a query band near the oracle-fingerprint manifold. *Fixed-size group querying* distributes the candidate pool across posterior operating points in the band while holding its total size fixed. *Posterior-aligned adaptation* tunes query reading while preserving the pretrained molecular prior.
- We establish new state-of-the-art exact-match accuracy and Morgan Tanimoto similarity on NPLIB1 and MassSpecGym. Our ablation study shows that all three components provide complementary gains. Candidate-pool scaling shows that efficient autoregressive molecular generation continues to improve recall with a little additional inference cost.

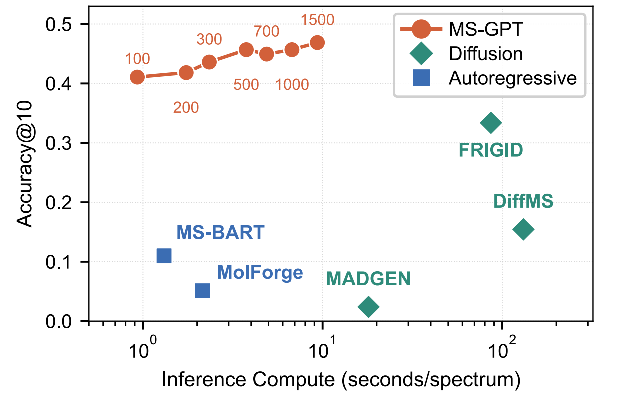

Figure 1: NPLIB1 Top-10 exact-match accuracy versus per-spectrum inference compute (log scale). The MS-GPT curve spans candidate-pool sizes B=100–1500; other markers denote diffusion-like and autoregressive baselines. Candidate-pool scaling steadily improves recovery with a near-linear increase in inference compute, placing MS-GPT on a favorable accuracy–efficiency frontier.

2 Related Work

2.1 Molecule-Language Models

Molecule-language models build on the serialization of molecular graphs as token sequences. SMILES provides a compact line notation for chemical structures [44], while later representations such as SELFIES and SAFE improve validity, substructure locality, and sequence generation [20, 29]. Building on these notations, Transformer [39] pretraining has produced transferable molecular representations for property prediction and representation learning, including SMILES Transformer, ChemBERTa, and MoLFormer [7, 15, 32]. Molecule strings have also supported cross-modal and generative models, including MolT5 for translating between molecules and text, GP-MoLFormer for scaled autoregressive

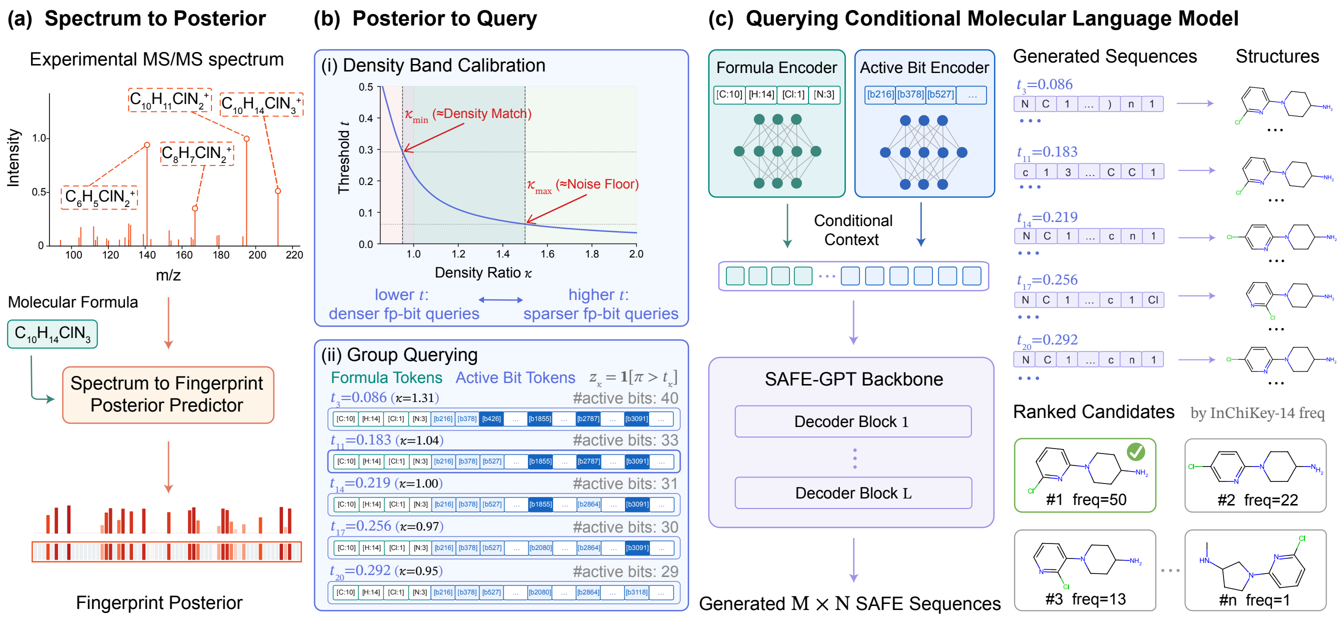

Figure 2: MS-GPT inference as spectrum-induced posterior querying. (a) A spectrum encoder maps an MS/MS spectrum and formula to a fingerprint posterior π. (b) Active-bit density band calibration maps π to a calibrated threshold interval, from which multiple posterior-derived queries are constructed. (c) The conditional molecule-language model generates SAFE sequences conditioned on each query; candidates are pooled across queries, grouped by InChIKey-14, and ranked by generation frequency.

generation, and SAFE-GPT for fragment-structured SAFE generation [11, 29, 33]. Together, these studies show that large-scale molecule-only corpora can endow sequence models with broad chemical priors.

Unlike prior molecule-language models for unconditional generation, property optimization, or text-conditioned design, our model conditionally maps molecular formulas and substructural fingerprints to compatible structures. This conditioning formulation enables molecule-only pretraining for MS/MS elucidation, but it also creates an oracle-to-posterior gap: pretraining conditions on oracle fingerprints, whereas inference supplies a noisy spectrum-induced fingerprint posterior. MS-GPT addresses this gap by making posterior-derived query construction the central mechanism linking spectral evidence to the pretrained molecular prior.

2.2 Structure Elucidation from Mass Spectrometry

Computational methods for MS/MS identification have traditionally relied on reference spectra or predefined candidate sets. Spectral-library search compares query spectra with measured references [1, 16, 41]; in silico fragmentation matches candidate-derived fragment masses to observed peaks [34], while spectral prediction methods rank candidates by matching predicted spectra to the query [40, 42]; CSI:FingerID and MIST predict molecular fingerprints for candidate ranking [10, 13]; and learned spectral embeddings support exact-match and molecular analogue search [5, 8, 18]. These approaches perform well when suitable references or candidates are available at inference time, but their search space is bounded by the chosen library or database. *De novo* structure elucidation relaxes this constraint by generating structures directly from spectra, thereby supporting the study of under-annotated natural products and the exploration of dark chemical space.

Some methods translate spectra into molecular strings [6, 22, 27, 35]. MSNovelist instead popularized a fingerprint-mediated route from fingerprint prediction to SMILES decoding [36]. This route is attractive because its decoder can be pretrained on molecule-only corpora, as demonstrated by Neuraldecipher and MolForge [21, 38]. Recent work has broadened the decoder family: MS-BART adopts sequence-to-sequence generation [14]; MADGEN combines scaffold retrieval with spectrum-guided generation [43]; DiffMS, MBGen, and FRIGID develop graph or masked diffusion decoders [2, 3, 37]; and flow-matching variants generate structures through iterative probability flows or formula-constrained molecular graphs [26, 28]. Orthogonal formulations explore test-time tuned language models and iterative optimization with forward spectral simulation [23, 24].

Across these lines of work, recent progress has largely centered on more expressive decoder architectures, while query construction from spectrum-induced fingerprint posteriors remains less explored. Fingerprint-mediated pipelines commonly collapse each posterior to a single thresholded query, discarding per-bit confidences and alternative structural hypotheses. MS-GPT applies active-bit density band calibration to form a band of posterior-derived queries near the oracle-fingerprint manifold, distributes a candidate pool of fixed size across the band, and ranks the pooled candidates by generation-frequency consensus. This posterior-query perspective complements stronger spectrum encoders and more expressive decoders: improved query construction can better align the spectrum-induced posterior with the conditional molecule-language model.

3 Method

3.1 Problem Formulation

Given a tandem mass spectrum x and its molecular formula C, *de novo* structure elucidation aims to generate the underlying molecular structure y directly, without a reference library or a predefined candidate set. Following the fingerprint-mediated paradigm, we introduce a discrete fingerprint variable $z \in \{0,1\}^d$, represented as a d-bit Morgan fingerprint [25, 31]:

$$p(y \mid x, C) = \mathbb{E}_{z \sim q_\phi(z|x,C)}\big[p_\theta(y \mid z, C)\big], \tag{1}$$

where q_ϕ is a spectrum encoder inducing a distribution over fingerprints and p_θ is a fingerprint-formula-conditioned decoder. For any molecule m, the corresponding oracle fingerprint is $z^\star_m = \mathrm{MorganFP}(m)$, allowing p_θ to be pretrained on molecule-only corpora without annotated spectra.

This decoupling creates a mismatch between pretraining and inference: the decoder is pretrained on oracle fingerprints, whereas at inference the oracle fingerprint is replaced by a *spectrum-induced fingerprint posterior* $\pi(x, C) \in [0,1]^d$, whose b-th entry estimates $q_\phi(z_b{=}1 \mid x, C)$ (Figure 2(a)). Existing pipelines collapse π into a single thresholded fingerprint, $\hat{z} = \mathbf{1}[\pi > t]$, and draw all candidates from $p_\theta(y \mid \hat{z}, C)$. This *point-query approximation* replaces the posterior expectation in Eq. (1) with one deterministic operating point, discarding the per-bit confidences and the alternative structural hypotheses they encode. MS-GPT instead separates posterior query construction into two complementary steps: active-bit density band calibration determines the query interval, while fixed-size group querying constructs M posterior queries at interpolated thresholds within the band and draws N samples per query. Across these queries, the conditional molecule-language model exploits learned fingerprint-bit co-occurrence to assemble fingerprint-encoded substructures into compatible molecular structures. In our implementation, a separate MIST spectrum encoder [13] instantiates q_ϕ for each spectral dataset, and p_θ is a fingerprint-formula-conditioned adaptation of SAFE-GPT [29] that generates molecular structures y as SAFE token sequences.

3.2 Conditional Molecule-Language Model

The conditional molecule-language model realizes $p_\theta(y \mid z, C)$ by generating SAFE sequence representations of molecular structures compatible with a fingerprint-formula query (Figure 2(c)). We pretrain the decoder to maximize the conditional likelihood of these sequences given oracle fingerprints and formulas,

$$\max_\theta \ \mathbb{E}_{m \sim \mathcal{D}}\big[\log p_\theta(y_m \mid z^\star_m, C_m)\big], \tag{2}$$

where each molecule m in the molecule-only corpus $\mathcal{D}$ provides a SAFE sequence y_m, an oracle fingerprint $z^\star_m$, and a formula C_m, all derived from its molecular structure. Because the model never observes spectra during pretraining, the decoder remains spectral-domain independent: a single molecule-only pretrained decoder can be shared unchanged across downstream spectral datasets, with spectral evidence entering only at query time through π.

Architecture. The conditional model adapts the SAFE-GPT decoder for dual fingerprint-formula conditioning, warm-started from its pretrained weights. A fingerprint encoder embeds the active bits of the fingerprint query z into a set of conditioning tokens, and a formula encoder embeds the per-element atom counts of C. Cross-attention modules interleaved through the transformer blocks then attend to this conditioning context. During pretraining, z is the oracle fingerprint $z^\star$; at inference, z is a posterior-derived thresholded query. SAFE is a fragment-based representation, so the decoder emits ring systems and functional groups as coherent units, aligning the generation target with the substructural semantics that the Morgan fingerprint encodes.

Formula-grouped pretraining. We group each mini-batch by exact molecular formula. Near-isomeric molecules that share a formula also share many fingerprint bits, placing structurally similar alternatives in direct contrast within a batch. This contrast encourages the model to resolve the remaining structural ambiguity under a fixed formula and oracle fingerprint, rather than exploiting that ambiguity as a generation shortcut (Figure 3(a)).

3.3 Posterior Query Construction

Active-bit density band calibration. The conditional model is trained only on oracle fingerprints, so it implicitly expects queries from the oracle-fingerprint manifold. However, the oracle fingerprint is unavailable at inference time, and the fingerprint must instead be obtained by thresholding the spectrum-induced posterior:

$$z(t) = \mathbf{1}[\pi > t].$$

The encoder's posterior π is a separately learned estimate that carries prediction error, so a naively thresholded query can move away from the oracle-fingerprint manifold: too low a threshold admits many weak bits, whereas too high a threshold discards plausible substructures. An appropriate threshold cannot be chosen example by example at inference time, since no oracle fingerprint is available for comparison. We therefore calibrate at the population level, matching the active-bit density of thresholded posterior queries to the ground-truth active-bit density estimated from the encoder's training annotations.

Let $\mathcal{T}_{\text{enc}}$ denote the spectrum encoder's training set and define the training-set active-bit density reference, measured by active-bit count, as

$$D_{\text{enc}} = \mathbb{E}_{(x,C,m) \sim \mathcal{T}_{\text{enc}}}\big[\|\mathrm{MorganFP}(m)\|_0\big], \tag{3}$$

where $\|\cdot\|_0$ counts active bits in a binary fingerprint. Thus D_{enc} estimates the ground-truth fingerprint density on the same annotated spectra that train the encoder and shape the posterior π. For a threshold t, define

$$A(t) = \mathbb{E}_{(x,C) \sim \mathcal{T}_{\text{enc}}}\big[\|\mathbf{1}[\pi(x,C) > t]\|_0\big] \tag{4}$$

as the expected active-bit count after thresholding the predicted posterior. Since $A(t)$ decreases with t, a target density ratio κ determines the threshold

$$t_\kappa = A^{-1}(\kappa D_{\text{enc}}), \tag{5}$$

so that $A(t_\kappa)/D_{\text{enc}} = \kappa$. Larger κ lowers the threshold and retains additional lower-confidence bits, whereas smaller κ raises the threshold and yields sparser queries. An active-bit density band is thus a density-ratio interval $\mathcal{K}$ around the oracle-fingerprint active-bit density. Through Eq. (5), this density band maps to a posterior-threshold interval; applying thresholds from this interval to an

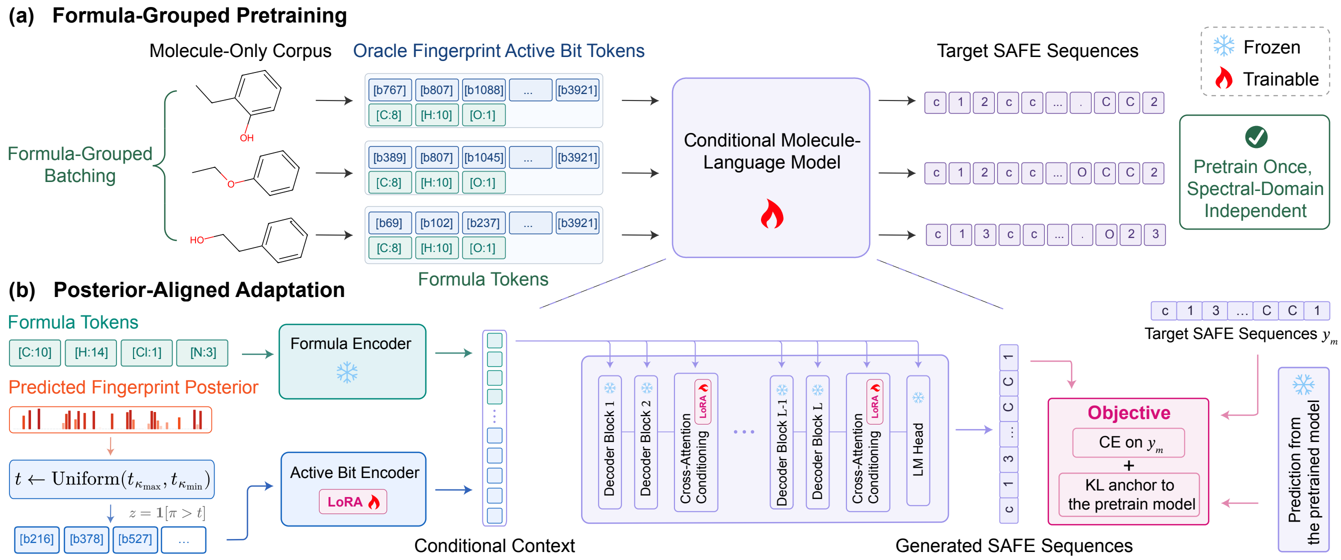

Figure 3: Pretraining and adaptation of the conditional molecule-language model. (a) Molecule-only formula-grouped pretraining teaches the model to map oracle fingerprint-formula queries to SAFE sequences. (b) Posterior-aligned adaptation trains lightweight LoRA modules on posterior-derived queries from the target spectral domain, while keeping the formula encoder, SAFE-GPT backbone, and language-model head frozen.

individual posterior produces a band of posterior-derived fingerprint queries with calibrated active-bit counts, keeping the queries near the oracle-fingerprint manifold in aggregate.

Fixed-size group querying. For candidate-pool size B, rather than drawing all candidates from a single thresholded query, we construct M posterior queries from the threshold interval induced by the calibrated active-bit density band $\mathcal{K} = [\kappa_{\min}, \kappa_{\max}]$ and draw N samples from each, giving $B = M \times N$ (Figure 2(b)). We invert Eq. (5) only at the two band endpoints,

$$t_{\kappa_{\max}} = A^{-1}(\kappa_{\max} D_{\mathrm{enc}}), \qquad t_{\kappa_{\min}} = A^{-1}(\kappa_{\min} D_{\mathrm{enc}}),$$

and linearly interpolate a prespecified number M of thresholds between them,

$$t_i = t_{\kappa_{\max}} + \frac{i-1}{M-1}(t_{\kappa_{\min}} - t_{\kappa_{\max}}), \quad i = 1, \ldots, M,$$

thereby approximating

$$p(y \mid x, C) \approx \frac{1}{M} \sum_{i=1}^{M} p_\theta(y \mid z_i, C), \qquad z_i = \mathbf{1}[\pi > t_i]. \tag{6}$$

for $M > 1$. We choose $\mathcal{K}$ to include density match (κ=1.00) and extend mainly toward denser queries. Appendix A.6 presents post hoc diagnostic results for this choice, while Appendix B.4 provides the density-band configuration details. The point-query approximation adopted by prior fingerprint-mediated pipelines is recovered as the degenerate case M=1.

3.4 Posterior-Aligned Adaptation

Density band calibration mitigates the aggregate mismatch between posterior-derived queries and the oracle-fingerprint manifold, but it does not resolve errors in which bits are active or how their dependencies differ from oracle fingerprints. Through molecule-only pretraining, the conditional model learns a molecular prior over fingerprint-bit co-occurrence: it can discount isolated active bits that conflict with the surrounding pattern and can still generate chemically coherent structures when some compatible substructures are missing from the query. However, if the spectrum encoder produces domain-specific posterior bias, thresholded queries may retain systematic bit-pattern shifts even inside the calibrated density band. These shifts affect how the pretrained decoder resolves competing chemically plausible completions. We therefore introduce a lightweight *posterior-aligned adapter* that aligns the query-reading pathway with posterior-derived active-bit patterns in the target spectral domain, while preserving the pretrained molecular prior.

Adapter architecture. Let $p_{\theta,\psi}$ denote the LoRA-adapted conditional model, where θ are the frozen pretrained weights and ψ are trainable low-rank adapter parameters [17]. We attach LoRA only to the query-reading pathway: the fingerprint encoder projections and the cross-attention modules that route fingerprint conditioning into the SAFE-GPT. The SAFE-GPT backbone, language-model head, and formula encoder remain frozen. This confines adaptation to how posterior-derived queries are encoded and routed into SAFE-token generation, rather than relearning the fingerprint-formula-conditioned molecular prior acquired during pretraining (Figure 3(b)).

Adaptation training. For each annotated pair (x, m) in the domain training set $\mathcal{T}_{\mathrm{pa}}$, we retain the predicted posterior $\pi(x, C)$, the oracle fingerprint $z_m^{\star}$, the formula C_m, and the SAFE target sequence y_m. During adaptation, the decoder input is not the oracle fingerprint. Instead, we sample a threshold t uniformly from the interval $[t_{\kappa_{\max}}, t_{\kappa_{\min}}]$ and form the thresholded query

$$z_t(x, C) = \mathbf{1}[\pi(x, C) > t].$$

The adapter is trained with teacher-forced SAFE likelihood on these posterior-derived queries, plus a frozen-teacher KL anchor:

$$\min_{\psi} \mathbb{E}_{\substack{(x,C,m)\sim\mathcal{T}_{\text{pa}} \\ t\sim\text{Uniform}([t_{\kappa_{\max}}, t_{\kappa_{\min}}])}} \Big[\alpha_g \, \mathcal{L}_{\text{CE}}(p_{\theta,\psi}; y_m, z_t, C_m) + \lambda_g \, \mathcal{L}_{\text{KL}}(p_\theta, p_{\theta,\psi}; y_m, z_t, C_m)\Big], \tag{7}$$

where $\mathcal{L}_{\text{CE}}$ is the next-token cross-entropy of y_m, and $\mathcal{L}_{\text{KL}}$ is a token-level KL from the frozen model to the adapted model on the same posterior-derived query. The weights (α_g, λ_g) depend on recoverability: the **frozen-hit** group, where the frozen model already generates the ground-truth molecular structure at the default candidate-pool size, receives weak CE and a strong KL anchor; the **residual** group, where the molecular structure is missed despite adequate posterior–oracle agreement, receives equal CE and KL weights to adapt the model to residual posterior bias; and the **low-coverage** group, where the molecular structure is missed with inadequate posterior–oracle agreement, receives no CE and only the KL anchor. Appendix B.3 specifies the recoverability grouping criterion and loss coefficients per group.

4 Experiments

4.1 Experimental Setup

Datasets and benchmarks. We evaluate under the known-formula protocol on two public MS/MS *de novo* benchmarks: **NPLIB1** [9] contains natural-product spectra from GNPS [41], and its training and test molecules are structurally similar, making it an in-distribution benchmark; **MassSpecGym** [4] follows an MCES-based structurally held-out split, measuring generalization to structurally distant molecules.

Evaluation metrics. Following prior work on known-formula MS/MS *de novo* evaluation, we report Top-k accuracy, Top-k Tanimoto similarity, and Top-k MCES distance over ranked generated candidates at $k \in \{1, 10\}$. **Top-k accuracy** measures whether any of the top-k candidates shares the same 2D structure (the first 14 InChIKey characters) with the ground-truth molecule. **Top-k Tanimoto similarity** measures the highest structural similarity to the ground-truth molecule among the top k candidates. **Top-k MCES distance** measures the smallest graph-edit distance to the ground-truth molecule among the top k candidates. For ablations, **Top-k % Close Match** (%CM) measures the percentage of spectra whose top-k generated candidates include a structure with RDKit topological-fingerprint Tanimoto similarity ≥ 0.675 [2]. Baseline MCES values are reported in the corresponding papers and reflect different evaluator configurations. For MS-GPT, we follow the DiffMS [3] configuration and summarize the differences in Appendix B.6.

Baselines. We compare against representative methods from the main decoder families for this task. Autoregressive sequence baselines include Spec2Mol [22], MIST+Neuraldecipher [21], MIST+MSNovelist [36], MS-BART [14], and MIST+MolForge [38]. Diffusion-like baselines include DiffMS [3], MBGen [37], and FRIGID [2]; scaffold-based baselines include MADGEN [43] and MSAnchor [30]. Baseline numbers are taken from their papers under the same known-formula protocol, except that the revised MIST+MolForge accuracies come from FRIGID [2].

Implementation. The MIST spectrum encoder trained on the corresponding dataset produces the fingerprint posterior π, and the lightweight LoRA adapter is trained on posterior-derived queries from that spectral domain. The underlying conditional molecule-language model is pretrained for two epochs on approximately 100M molecular structures, after excluding connectivity-equivalent structures from the validation and test splits of both benchmarks. Unless stated otherwise, evaluation uses candidate-pool size B=100, with M=20 query groups and N=5 samples per group. Appendix A.7 reports additional results on pretraining data scaling, while Appendix B provides full implementation details.

4.2 Main Results

Table 1 summarizes performance. MS-GPT achieves the highest exact-match accuracy on both benchmarks, reaching Top-1/Top-10 accuracies of 29.76%/41.07% on NPLIB1 and 23.91%/28.65% on MassSpecGym. Compared with FRIGID, the strongest exact-match baseline, MS-GPT improves Top-1/Top-10 by 4.73/7.70 percentage points on NPLIB1 and 5.62/6.65 percentage points on MassSpecGym. These gains hold on both the in-distribution NPLIB1 benchmark and the structurally distant MassSpecGym benchmark. MS-GPT shares the same conditional molecule-language model across benchmarks; only the MIST spectrum encoder and lightweight LoRA adapter are trained for each benchmark. MCES offers a complementary view that MS-GPT improves over all autoregressive baselines and records the best Top-10 MCES on NPLIB1, while diffusion-like decoders lead on the remaining MCES comparisons. This pattern is consistent with the generative trade-off: iterative diffusion can refine local graph structure, whereas autoregressive molecular generation scales candidate pools efficiently.

4.3 Component Ablation

Table 2 reports a component ablation of MS-GPT on NPLIB1. Removing posterior-aligned adaptation gives MS-GPT-Base. Relative to MS-GPT-Base, MS-GPT improves Top-1/Top-10 accuracy by 1.34/7.29 points. The gains are concentrated at Top-10, indicating that posterior-aligned adaptation mainly improves candidate-pool recall. Starting from MS-GPT-Base, we then isolate the two query-construction components. Replacing active-bit density band calibration with randomly sampled queries outside the calibrated band reduces Top-1/Top-10 accuracy by 3.87/2.23 points. Collapsing fixed-size group querying to a single threshold gives three point-query controls: a density-matched operating point at κ=1.00 and two operating points whose thresholds are selected on the validation set to maximize F_1 and cosine similarity, respectively. Even the strongest point-query control falls below MS-GPT-Base by 4.02/4.91 Top-1/Top-10 points. The MassSpecGym counterpart in Appendix A.1 shows the same pattern.

4.4 Posterior-Fidelity Analysis

We stratify NPLIB1 by spectrum-induced posterior fidelity to investigate how much performance depends on posterior quality. For each test spectrum, posterior fidelity T_c is the largest Tanimoto

Table 1: *De novo* structural elucidation performance on NPLIB1 [9] and MassSpecGym [4], assuming a known chemical formula. The best result in each column is bolded, and methods are ordered by Top-1 accuracy within each benchmark. Baseline entries are taken from the respective papers, with † indicating revised MIST+MolForge accuracies reported by FRIGID [2]. ‡ Baseline MCES evaluators have inconsistent configurations; Appendix B.6 summarizes these differences.

	Top-1			Top-10		
Model	Accuracy ↑	MCES‡ ↓	Tanimoto ↑	Accuracy ↑	MCES‡ ↓	Tanimoto ↑
NPLIB1						
Spec2Mol [22]	0.00	48.57	0.12	0.00	17.58	0.16
MADGEN [43]	2.10	20.56	0.22	2.39	12.69	0.27
MIST+MolForge† [38]	2.24	14.16	0.40	5.11	10.96	0.47
MIST+Neuraldecipher [21]	2.32	10.26	0.35	6.11	8.53	0.43
MIST+MSNovelist [36]	5.40	13.10	0.34	11.04	8.57	0.44
MS-BART [14]	7.45	9.66	0.44	10.99	8.31	0.51
DiffMS [3]	8.34	11.95	0.35	15.44	9.23	0.47
MSAnchor [30]	8.51	11.12	0.38	16.90	8.95	0.49
MBGen [37]	12.20	7.72	0.41	22.29	6.71	0.50
FRIGID [2]	25.03	**7.10**	0.58	33.37	5.56	0.64
MS-GPT (ours)	**29.76**	7.38	**0.61**	**41.07**	**5.51**	**0.67**
MassSpecGym						
MIST+MSNovelist [36]	0.00	39.84	0.06	0.00	18.83	0.15
Spec2Mol [22]	0.00	36.78	0.12	0.00	36.02	0.16
MIST+Neuraldecipher [21]	0.00	22.93	0.14	0.00	21.76	0.16
MS-BART [14]	1.07	16.47	0.23	1.11	15.12	0.28
MADGEN [43]	1.31	27.47	0.20	1.54	16.84	0.26
DiffMS [3]	2.30	18.45	0.28	4.25	14.73	0.39
MSAnchor [30]	2.68	16.57	0.32	4.67	14.12	0.41
MBGen [37]	7.58	**13.25**	0.38	12.54	**10.16**	0.41
MIST+MolForge† [38]	10.73	22.15	0.37	14.48	17.88	0.41
FRIGID [2]	18.29	13.49	0.43	22.00	11.65	0.47
MS-GPT (ours)	**23.91**	15.50	**0.52**	**28.65**	11.74	**0.54**

Table 2: Component ablation of MS-GPT on NPLIB1. Cells report Top-1/Top-10.

Variant	Accuracy ↑	Tanimoto ↑	%CM ↑
MS-GPT	**29.76/41.07**	**0.61/0.67**	**63.99/70.54**
w/o Posterior-Aligned Adaptation			
MS-GPT-Base	28.42/33.78	0.58/0.62	62.05/66.37
w/o Posterior-Aligned Adaptation, Density Band Calibration			
Outer-band querying	24.55/31.55	0.54/0.57	54.61/59.67
w/o Posterior-Aligned Adaptation, Group Querying			
Density match (κ=1.00)	24.40/28.87	0.52/0.54	54.91/57.74
F_1 (t=0.340)	22.47/25.30	0.49/0.51	50.45/53.12
Cosine (t=0.377)	21.88/24.55	0.48/0.49	50.15/51.93

similarity between the oracle fingerprint and any posterior-derived query over the calibrated density band. This yields low (T_c<0.5), middle (0.5≤T_c<0.7), and high (T_c≥0.7) bins [12, 19]. Figure 4 shows that MS-GPT accuracy rises sharply from the low-fidelity bin to the high-fidelity bin; the MassSpecGym counterpart in Appendix A.2 shows an even more pronounced trend. This benchmark-spanning dependence confirms that the fingerprint condition steers molecular generation: the model is not simply drawing plausible structures from the pretrained molecular prior independently of spectral evidence. Relative to MS-GPT-Base, posterior-aligned adaptation contributes most in the middle and high bins, especially at Top-10. The low-fidelity bin remains difficult for both models, identifying the spectrum-induced posterior as a central remaining bottleneck.

4.5 Pretraining-Proximity Analysis

We stratify NPLIB1 by pretraining proximity to investigate whether performance is driven mainly by close analogs in the fingerprint-formula pretraining set. Pretraining proximity is the maximum Tanimoto similarity between each test molecule and any pretraining molecule with the same formula. The same cutoffs as in Section 4.4 define low, middle, and high proximity bins. Figure 5 shows that accuracy improves from the low-proximity bin to the middle bin, indicating that nearby structures provide a helpful conditional prior.

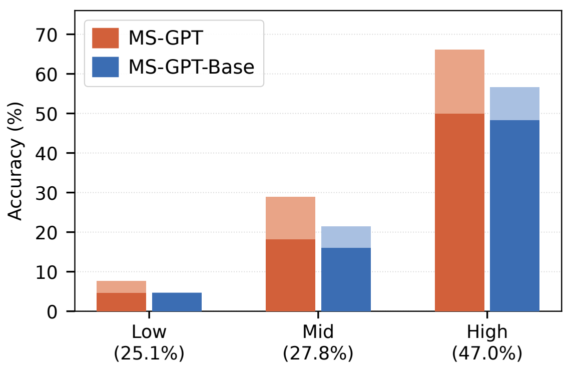

Figure 4: NPLIB1 Top-1/Top-10 exact-match accuracy by posterior-fidelity bin. Parentheses indicate test-set proportions; dark segments show Top-1 accuracy and total bar heights show Top-10 accuracy.

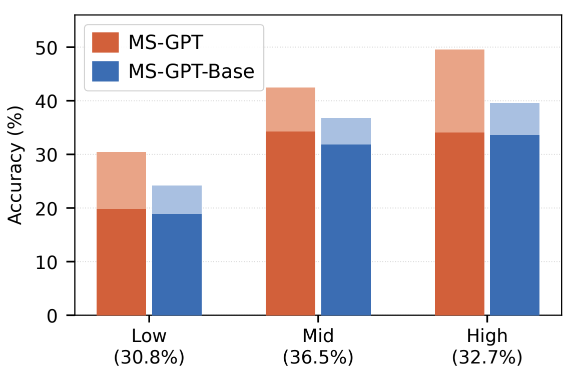

Figure 5: NPLIB1 Top-1/Top-10 exact-match accuracy by pretraining-proximity bin. Parentheses indicate test-set proportions; dark segments show Top-1 accuracy and total bar heights show Top-10 accuracy.

However, the trend changes only mildly from the middle to high bin, and the gain of MS-GPT over MS-GPT-Base is not concentrated in the high-proximity bin. The result therefore suggests that performance benefits from nearby pretraining neighborhoods but is not primarily driven by nearest-neighbor memorization. This behavior is consistent with the molecule-language model learning reusable co-occurrence patterns between active-bit conditioning tokens and SAFE tokens, rather than relying on whole-molecule structural similarity. This distinction separates reusable substructure-level transfer from retrieval driven by close whole-molecule analogs in the pretraining corpus. The corresponding MassSpecGym analysis in Appendix A.3 shows the same conclusion.

4.6 Inference Efficiency and Scaling

MS-GPT generates each candidate in a single autoregressive pass, avoiding iterative graph generation or refinement. Figure 1 compares per-spectrum inference compute with Top-10 exact-match accuracy on NPLIB1. MS-GPT occupies a favorable accuracy–cost region: diffusion-like decoders require one to two orders of magnitude more time, whereas autoregressive baselines are comparably fast but far less accurate. Posterior querying therefore improves accuracy without giving up the efficiency of autoregressive molecular generation.

Autoregressive molecule generation also makes the candidate pool inexpensive to enlarge. In the candidate-pool scaling experiment, we keep the query grid fixed and draw more samples from each query, expanding the pooled candidate set with a near-linear cost increase. Figure 1 shows that Top-10 exact-match accuracy continues to improve as the candidate-pool size grows. The MassSpecGym counterpart in Appendix A.4 shows the same trend, and Appendix A.5 examines how a fixed candidate-pool size is allocated over query groups. Detailed case studies are provided in Appendix C.

5 Conclusion

We introduced MS-GPT, a spectrum-induced posterior-querying framework for fingerprint-mediated MS/MS *de novo* structure elucidation. Instead of collapsing the fingerprint posterior to one point estimate, MS-GPT converts it into a calibrated band of posterior-derived queries near the oracle-fingerprint manifold and queries a conditional molecule-language model pretrained on a molecule-only corpus. Active-bit density band calibration, fixed-size group querying, and posterior-aligned adaptation bridge the oracle-to-posterior gap while preserving the pretrained molecular prior. On NPLIB1 and MassSpecGym, MS-GPT sets a new state of the art in exact-match accuracy and Morgan Tanimoto similarity; ablations show complementary gains from all three components. Candidate-pool scaling further improves Top-10 recovery with near-linear growth in inference compute. These results frame MS/MS *de novo* elucidation as posterior querying over molecule-language models, with future progress depending on more faithful spectrum-induced posteriors and stronger conditional molecular generation.

6 Limitations and Ethical Considerations

Our study is limited to two public MS/MS benchmarks under the known-formula setting and depends on a fixed-dimensional fingerprint posterior predicted by the spectrum encoder; missing or incorrect posterior information can therefore affect candidate generation. The datasets contain no human participants or private data, so informed consent and institutional ethics review are not applicable. Practical applications, such as compound identification in drug discovery and environmental monitoring, require expert review and experimental validation of the predicted structures.

7 Generative AI Usage

Generative AI tools assisted with language polishing of the authors' original draft and code debugging. All suggested textual revisions and code changes were reviewed and verified by the authors.

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A Additional Experimental Results

A.1 Component Ablation

Table 3 presents the MassSpecGym counterpart to the NPLIB1 ablation in Table 2. MS-GPT leads consistently in exact-match accuracy, structural similarity, and close-match rate. Removing posterior-aligned adaptation causes a broad but moderate decline, indicating that adaptation helps the pretrained generator respond to posterior-derived conditions. Query construction has a stronger effect: sampling outside the calibrated density band weakens performance, while every single-threshold control falls short of grouped querying. The density-matched query is the strongest of these point controls, but it does not recover the benefit of covering multiple posterior conditions. This pattern mirrors NPLIB1 and shows that density calibration, group coverage, and posterior-aligned adaptation remain complementary in the more structurally shifted setting.

Table 3: Component ablation of MS-GPT on MassSpecGym. Cells report Top-1/Top-10.

Variant	Accuracy ↑	Tanimoto ↑	%CM ↑
MS-GPT	**23.91/28.65**	**0.52/0.54**	**52.51/55.27**
w/o Posterior-Aligned Adaptation			
MS-GPT-Base	22.77/26.96	0.50/0.52	50.83/53.33
w/o Posterior-Aligned Adaptation, Density Band Calibration			
Outer-band querying	19.70/22.84	0.41/0.43	43.91/45.14
w/o Posterior-Aligned Adaptation, Group Querying			
Density match (κ=1.00)	20.08/22.80	0.44/0.46	45.47/47.23
F_1 (t=0.257)	19.58/22.78	0.43/0.44	44.17/45.85
Cosine (t=0.269)	19.42/22.56	0.42/0.44	43.91/45.55

A.2 Posterior-Fidelity Analysis

Figure 6 extends the posterior-fidelity analysis to MassSpecGym. Accuracy separates sharply across the three bins: both variants recover almost no exact matches in the low-fidelity group, improve in the middle group, and perform substantially better in the high-fidelity group. The shared trend of MS-GPT and MS-GPT-Base shows that the quality of the spectrum-induced posterior remains a primary determinant of recovery under structural shift. The advantage of MS-GPT emerges mainly once the posterior provides informative structural evidence and is clearest for Top-10 recovery in the high-fidelity group. Posterior-aligned adaptation therefore helps translate informative queries into broader candidate-pool recovery, but does not compensate for poorly grounded queries. The low-fidelity group remains the main target for improving the spectrum encoder and posterior construction.

A.3 Pretraining-Proximity Analysis

Figure 7 extends the pretraining-proximity analysis to MassSpecGym. Both MS-GPT and MS-GPT-Base peak in the middle-proximity group and decline in the high-proximity group, rather than improving monotonically with proximity. MassSpecGym is structurally held out, and connectivity-equivalent evaluation structures are

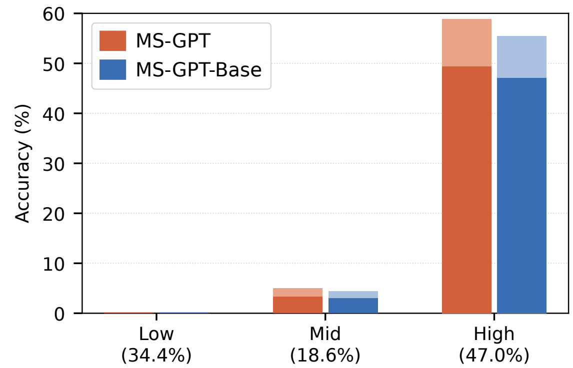

Figure 6: MassSpecGym Top-1/Top-10 exact-match accuracy by posterior-fidelity bin. Parentheses indicate test-set proportions; dark segments show Top-1 accuracy and total bar heights show Top-10 accuracy.

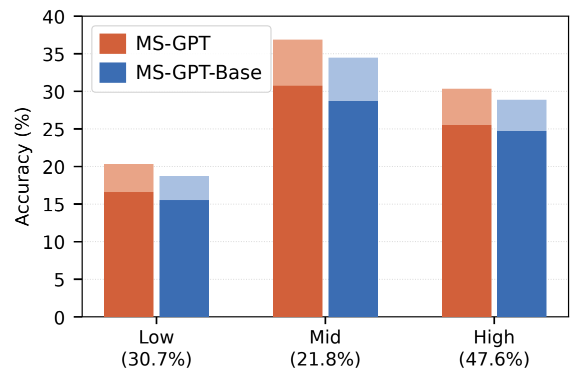

Figure 7: MassSpecGym Top-1/Top-10 exact-match accuracy by pretraining-proximity bin. Parentheses indicate test-set proportions; dark segments show Top-1 accuracy and total bar heights show Top-10 accuracy.

additionally excluded from the corpus for fingerprint–formula conditional pretraining. A high-proximity neighbor is therefore structurally similar but cannot be the exact target. The decline is consistent with competition from familiar same-formula isomers. Alongside the monotonic posterior-fidelity trend in Appendix A.2, this contrast suggests that posterior quality shapes whether the pretrained neighborhood supports recovery or supplies convincing decoys. An informative spectrum-induced posterior distinguishes the target from close alternatives, whereas weak evidence leaves the learned conditional prior insufficiently constrained. MS-GPT remains ahead across all three groups, suggesting that posterior-aligned adaptation mitigates this conflict without eliminating it.

A.4 Inference Efficiency and Scaling

Figure 8 plots MassSpecGym Top-10 exact-match accuracy against per-spectrum inference compute; Table 4 reports the corresponding fixed-grid sweeps for NPLIB1 and MassSpecGym. Across both sweeps, Top-10 recovery generally rises as the candidate pool grows,

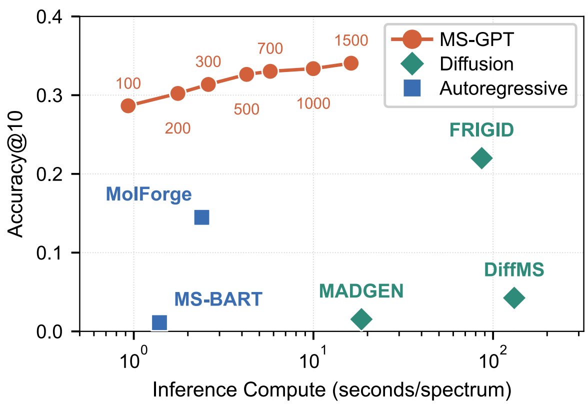

Figure 8: MassSpecGym Top-10 exact-match accuracy versus per-spectrum inference compute (log scale). The MS-GPT curve spans candidate-pool sizes B=100–1500; other markers denote baseline methods.

Table 4: Candidate-pool scaling of MS-GPT on NPLIB1 and MassSpecGym. Rows report per-spectrum compute on a single NVIDIA RTX 6000 Ada GPU and Top-1/Top-10 exact-match accuracy (%) at each candidate-pool size B.

B	Compute (s/spec)↓	Top-1↑	Top-10↑
	NPLIB1		
100	0.93	29.76	41.07
200	1.74	29.76	41.82
300	2.34	31.40	43.60
500	3.76	31.10	45.68
700	4.88	31.10	44.94
1000	6.75	30.80	45.68
1500	9.35	30.95	46.88
	MassSpecGym		
100	0.93	23.91	28.65
200	1.76	24.42	30.22
300	2.60	24.60	31.36
500	4.25	25.02	32.64
700	5.75	25.12	33.03
1000	9.98	25.10	33.37
1500	16.21	25.40	34.06

whereas Top-1 changes more modestly and shows small fluctuations. Together, these trends indicate that additional sampling broadens candidate coverage more reliably than it alters the leading prediction. Although the gains taper at larger candidate-pool sizes, MS-GPT still attains strong Top-10 exact-match accuracy on MassSpecGym at comparatively low per-spectrum inference compute.

A.5 Allocation at Fixed Candidate-Pool Size

Figures 9 and 10 examine how a fixed candidate-pool size is divided between query diversity and per-query sampling depth. On both

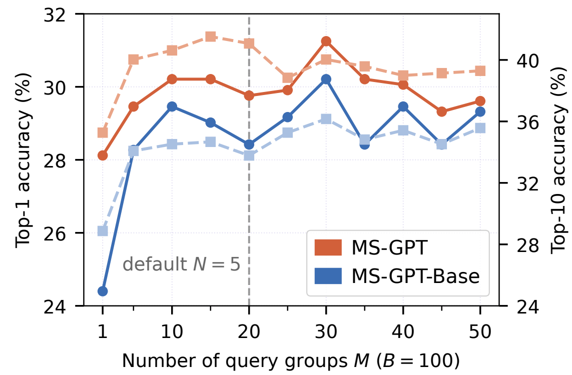

Figure 9: NPLIB1 Top-1/Top-10 exact-match accuracy across query-group allocations at fixed candidate-pool size B=100. Increasing M refines the grid over the calibrated density band while reducing samples per group (N≈100/M). Solid circles and dashed squares denote Top-1 and Top-10 accuracy on the left and right axes, respectively; the vertical line marks the default M=20, N=5.

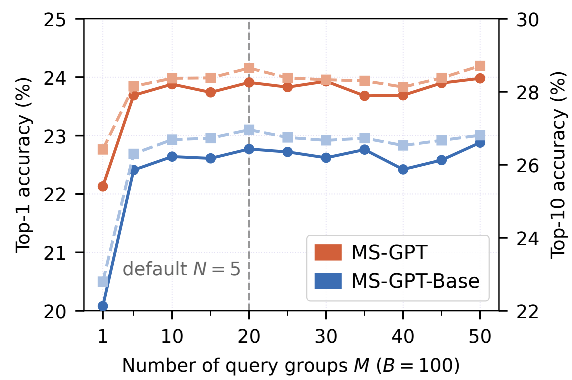

Figure 10: MassSpecGym Top-1/Top-10 exact-match accuracy across query-group allocations at fixed candidate-pool size B=100. Increasing M refines the grid over the calibrated density band while reducing samples per group (N≈100/M). Solid circles and dashed squares denote Top-1 and Top-10 accuracy on the left and right axes, respectively; the vertical line marks the default M=20, N=5.

datasets, the clearest improvement appears when the single-point query is replaced by a small grid of posterior queries. Performance then remains broadly stable as the grid is refined, with excess generations trimmed where necessary to preserve B: MassSpecGym exhibits a particularly flat plateau, whereas NPLIB1 fluctuates more without a consistent trend toward larger M. This pattern indicates that the main benefit comes from spanning plausible posterior operating points rather than repeatedly sampling from a single threshold; once multiple queries are included, the precise allocation matters less. The default M=20, N=5 balances coverage across the query band with per-query sampling depth and lies within this stable regime. MS-GPT remains ahead of MS-GPT-Base throughout both sweeps, indicating that the advantage of posterior-aligned adaptation persists across candidate-pool allocations.

A.6 Density-Ratio Diagnostics

Appendix B.4 specifies the shared endpoints $\kappa_{\min}$ and $\kappa_{\max}$, chosen empirically with reference to validation-set metrics. The resulting band is intentionally conservative, beginning near density match and extending toward denser queries to retain a wider set of lower-confidence bits. This choice was guided by the expectation that co-occurrence patterns learned between active-bit conditioning tokens and SAFE tokens during conditional pretraining could help the model tolerate isolated noisy bits while benefiting from the additional structural cues retained by denser queries.

Figures 11 and 12 provide a post hoc analysis of this calibration. Across the sweep, increasing κ produces denser single-threshold queries, as confirmed by the monotonic active-bit curves. The high-fidelity share peaks near density match; the middle-fidelity share remains elevated farther into the denser regime; and the low-fidelity share increases toward both extremes. To our surprise, validation and test exhibit similar κ-dependent patterns throughout the sweep. The curves nearly overlap on MassSpecGym; on NPLIB1, the absolute fidelity shares differ more, but the same qualitative progression is preserved. This cross-split agreement suggests that κ provides a consistent coordinate for posterior fidelity across the observed splits. The position of the high-fidelity peak and its cross-split stability suggest that the selected band may have been more conservative than necessary; a modest shift toward smaller κ would filter more low-confidence bits and move the query grid toward the high-fidelity region, potentially improving Top-1 exact-match accuracy.

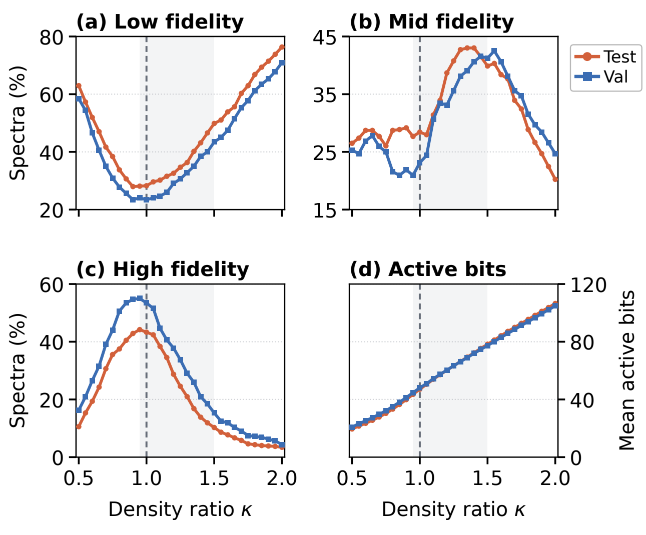

Figure 11: NPLIB1 posterior-fidelity bin proportions and mean active-query-bit count versus density ratio κ. Curves compare validation and test splits; the dashed line marks density match (κ=1.00), and shading marks the calibrated density band $\mathcal{K} = [0.95, 1.50]$.

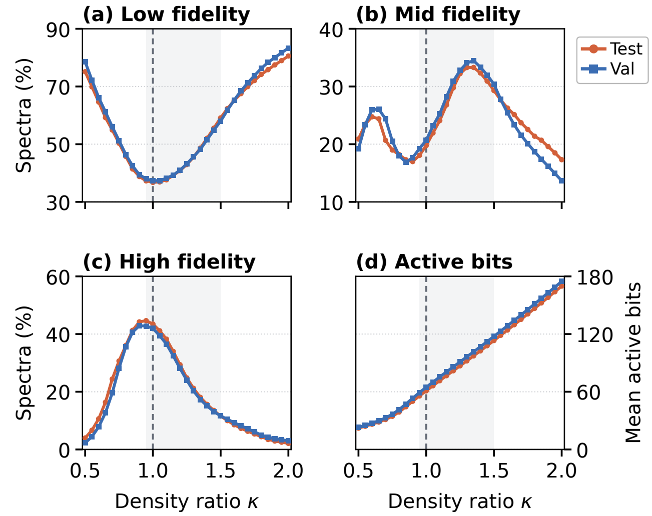

Figure 12: MassSpecGym posterior-fidelity bin proportions and mean active-query-bit count versus density ratio κ. Curves compare validation and test splits; the dashed line marks density match (κ=1.00), and shading marks the calibrated density band $\mathcal{K} = [0.95, 1.50]$.

A.7 Pretraining Data Scaling

Table 5 examines a separate one-epoch pretraining trajectory of MS-GPT-Base on a filtered, approximately 1B-structure SAFE-GPT corpus, with fingerprint–formula conditioning retained throughout. The architecture, candidate-pool size, and density-band inference protocol remain fixed, and connectivity-equivalent validation and test structures from both benchmarks are excluded. Its 100M checkpoint is therefore distinct from the main two-epoch model. Exact-match accuracy and Morgan Tanimoto similarity improve throughout the sweep, indicating that greater pretraining exposure strengthens both the leading prediction and the broader candidate pool. Gains become more gradual at larger scales but persist through the largest checkpoint, suggesting diminishing returns without clear saturation.

Table 5: Pretraining data scaling of MS-GPT-Base on NPLIB1. Rows list one-epoch checkpoints by cumulative structures seen. Accuracy and Tanimoto report Top-1/Top-10 exact-match accuracy (%) and maximum Morgan Tanimoto similarity, respectively; cumulative GPU-hours are estimated across eight NVIDIA A800 GPUs. Bold marks the best result.

Structures Seen	GPU Hours	Accuracy ↑	Tanimoto ↑
5M	7.8	12.05/16.52	0.325/0.352
10M	14.9	16.96/21.28	0.393/0.418
50M	71.8	24.85/31.10	0.513/0.546
100M	143.1	26.04/33.48	0.526/0.567
500M	712.0	30.65/36.46	0.555/0.583
1B	1422.5	**31.55/37.80**	**0.577/0.609**

B Experimental Details

B.1 Model Architecture

Our conditional molecule-language model augments the SAFE-GPT decoder [29] with fingerprint and formula conditioning through cross-attention (Figure 2(c)). The model contains 150.1M unique parameters: an 86.7M GPT-2 backbone warm-started from SAFE-GPT and 63.4M parameters in the fingerprint encoder, formula encoder, cross-attention blocks, and shared context projection, all initialized from scratch. Table 6 summarizes the architecture and parameter counts.

Backbone. The decoder is a 12-layer GPT-2-style transformer with model width 768, 12 attention heads, and a maximum sequence length of 256 tokens. Its vocabulary extends the 1880-token SAFE-GPT vocabulary with the tokens < and >, giving 1882 tokens in total. The language-model head shares its weight matrix with the input-token embedding. To warm-start the backbone, we copy the shared token-embedding rows, all transformer-block weights, the final layer norm, and the first 256 positional-embedding rows from the public `datamol-io/safe-gpt` checkpoint; the two added token rows are initialized from scratch.

Fingerprint encoder. We represent each query as a 4096-bit radius-2 Morgan fingerprint [25, 31] and retain only its active bits. Each selected bit is represented by a learned embedding of its index at width 768. A 2-layer pre-norm Transformer encoder (12 heads, feed-forward width 3072, GELU) then contextualizes the per-bit embeddings. We retain at most 256 active bits in ascending bit-index order and pad shorter sequences to this length with an accompanying validity mask. Thus, the fingerprint encoder supplies up to 256 valid conditioning tokens. During conditional pretraining, the fingerprint input is the oracle fingerprint $z^\star$ computed from the molecule; at inference, it is a thresholded posterior query $z = \mathbf{1}[\pi > t]$.

Formula encoder. We represent the molecular formula as integer atom counts over 14 elements (C, H, N, O, S, P, F, Cl, Br, I, B, Si, Se, As). For each element, a learned element embedding is concatenated with a two-layer MLP encoding of its atom count, projected to width 768, and layer-normalized. This produces 14 formula-conditioning tokens.

Cross-attention conditioning. The padded fingerprint sequence and 14 formula tokens are concatenated, together with their validity mask, into a shared context of 270 token positions and passed through one shared linear projection. A conditioning block is inserted after decoder layers 2, 4, 6, 8, 10, and 12. Each block applies pre-norm cross-attention from the decoder hidden states to the shared context, followed by a pre-norm feed-forward sublayer ($768 \rightarrow 3072 \rightarrow 768$, GELU); residual connections wrap both sublayers. Fingerprint and formula information therefore enter through one shared conditioning pathway, which remains present for every training example.

B.2 Conditional Pretraining

Starting from the SAFE-GPT initialization described above, we train the fingerprint–formula-conditioned model with the molecule-only objective in Eq. (2); no spectra enter this stage. The conditionally pretrained decoder is shared across spectral datasets; within the decoder, only the lightweight adapter is trained separately for each dataset.

Corpus construction. For the main model, we construct the corpus from a 117,280,040-record subset of the SAFE-GPT molecule collection. Each SAFE string is converted to SMILES; stereochemistry is removed before canonicalization, fingerprint computation, and formula extraction. We discard 17,220,290 records that fail SAFE-to-SMILES conversion, RDKit parsing, or feature computation, and further exclude 52,391 records whose 14-character InChIKey connectivity block matches one of the 9,413 unique blocks in the validation or test split of either benchmark. The resulting main corpus contains 100,007,359 records, with connectivity-equivalent validation and test structures excluded from conditional pretraining.

Formula-grouped batching. We organize batches around exact molecular formula (Figure 3(a)). Formula groups containing at least 64 records, the per-GPU batch size, are shuffled internally and partitioned into complete same-formula batches. Smaller groups are pooled, shuffled, and partitioned into mixed-formula batches. Only complete local batches are retained; the combined batch list is shuffled each epoch and divided evenly among the eight data-parallel processes. This policy preserves the likelihood objective while increasing the within-batch concentration of alternatives that share the same formula.

Optimization. We optimize all 150.1M parameters for 2 epochs ($\approx$ 195k optimizer updates) with AdamW ($\beta = (0.9, 0.95)$, weight decay 0.01). The learning rate warms up for 2000 updates to a peak of 3×10^{-4} and then follows a cosine decay. Training employs bfloat16 mixed precision, gradient clipping at norm 1.0, and a maximum sequence length of 256. The effective global batch size is 1024 (64 per GPU, eight GPUs, and two gradient-accumulation steps). The main two-epoch run requires approximately 358 aggregate GPU-hours across eight NVIDIA A800 GPUs.

B.3 Posterior-Aligned Adaptation

For each spectral dataset, we fit a separate posterior-aligned adapter to the shared conditional molecule-language model (Section 3.4, Figure 3(b)).

Adapter placement. We attach rank-4 LoRA modules [17] ($\alpha = 8$, dropout 0.05) to 39 linear layers: the fingerprint encoder's value and combine projections, the shared context projection, and six projections in each cross-attention block (query, key, value, output, and two feed-forward projections). All 150.1M pretrained weights remain frozen; the 350,212 trainable LoRA parameters constitute 0.23% of the model. Before attaching LoRA to the query, key, and value projections, we split each fused query–key–value projection and copy the corresponding pretrained weight slices, leaving the model's initial function unchanged.

Recoverability grouping. The oracle fingerprint available for each annotated training pair serves only to assign its recoverability group. Under the default B=100 posterior-querying protocol, a pair is **frozen-hit** if the frozen candidate pool contains an InChIKey-14 match. For a miss, we compute the posterior–oracle agreement $c = \max_{t \in \mathcal{G}} \mathrm{Tanimoto}(\mathbf{1}[\pi > t], z^\star)$, over the same dataset-specific 20-point density-band grid $\mathcal{G}$ applied at inference (Appendix B.4).

Table 6: Architecture and unique parameter counts of the conditional molecule-language model. The GPT-2 backbone is warm-started from SAFE-GPT, while the conditioning components are initialized from scratch. The tied token embedding and language-model head are counted once. Dropout is 0.1 in the GPT-2 backbone, fingerprint Transformer, and cross-attention blocks.

Component	Configuration	#Params
Backbone (GPT-2)	12 layers, d=768, 12 heads, sequence length 256, vocabulary 1882	86.7M
Fingerprint encoder	4096-bit radius-2 Morgan, $\leq$ 256 active bits, 2-layer Transformer	18.5M
Formula encoder	14 elements, count MLP	1.8M
Cross-attention	after layers 2, 4, 6, 8, 10, 12; 12 heads; FFN 3072	42.5M
Context projection	Linear 768 $\rightarrow$ 768	0.6M
Total		150.1M

Table 7: Recoverability groups and per-example loss weights for posterior-aligned adaptation (Eq. (7)). A frozen hit is an InChIKey-14 match among the valid, formula-consistent outputs in the default B=100 pool; c is the maximum posterior–oracle Tanimoto over the dataset-specific 20-point density-band grid.

Group	Criterion	CE α_g	KL λ_g
frozen-hit	target in frozen B=100 pool	0.05	2.0
residual	miss and $c \geq 0.40$	1.0	1.0
low-coverage	miss and $c < 0.40$	0.0	0.5

Misses with $c \geq 0.40$ are **residual**; the remainder are **low-coverage**. This grouping applies only during adaptation. The CE weight is largest for residual cases, small for frozen hits, and zero for low-coverage cases; the KL weight is nonzero for all three groups and largest for frozen hits (Table 7).

Objective and optimization. For each example, we draw t uniformly from its dataset-specific density-band interval and form $z_t = \mathbf{1}[\pi > t]$. Equation (7) combines group-weighted teacher-forced SAFE cross-entropy with a forward-KL anchor from the frozen teacher to the adapted student; both receive the same z_t, and the KL is averaged over non-padding sequence positions. We optimize the LoRA parameters with AdamW ($\beta = (0.9, 0.95)$, weight decay 0.01), learning rate 10^{-4}, 150 warmup steps followed by cosine decay, bfloat16 precision, gradient clipping at norm 1.0, and maximum sequence length 256. The global batch size is 64 (eight examples on each of eight NVIDIA A800 GPUs), without gradient accumulation. Training runs for 3000 optimizer steps on NPLIB1 and 4000 on MassSpecGym. Based on held-out validation performance, we select the checkpoints at steps 3000 and 2500, respectively.

B.4 Density-Band Configuration

We apply a shared active-bit density-ratio band $\mathcal{K} = [0.95, 1.50]$ to both benchmarks. The endpoints were chosen empirically with reference to validation-set metrics. The interval contains the density-match point (κ=1.00), extends mainly toward denser queries, and reaches only slightly into the sparse regime. Extending farther below κ=1.00 would map to the flat, high-threshold region of $A(t)$, where small errors in the estimated active-bit count can produce large threshold shifts. We therefore share the band in density-ratio space while deriving the corresponding thresholds from each encoder's training split.

We estimate D_{enc} and $A(t)$ from the corresponding encoder training split alone, obtaining D_{enc} = 47.95 active bits for NPLIB1 and 42.03 for MassSpecGym. Inverting the endpoints of $\mathcal{K}$ through Eq. (5) gives $(t_{1.50}, t_{0.95}) = (0.062, 0.292)$ and $(0.069, 0.274)$, respectively. At inference, we linearly interpolate M=20 thresholds between these endpoints to form 20 query groups and draw N=5 sequences per group. Thus, the default candidate-pool size $B = M \times N = 100$ denotes 100 decoder samples per spectrum before candidates are grouped by InChIKey-14 and ranked by generation frequency.

B.5 Inference and Compute Benchmarking

Posterior and generation. For each benchmark, the corresponding dataset-specific MIST [13] checkpoint released by FRIGID [2] produces π and remains frozen throughout our experiments. Given this posterior, MS-GPT constructs the dataset-specific threshold grid and samples SAFE sequences with temperature 1.0, top-k 50, and top-p 0.95. Candidates are grouped by InChIKey-14 and ranked by generation frequency, with posterior similarity breaking ties. The default candidate-pool size is B=100 decoder samples per spectrum.

Compute benchmarking. We measure per-spectrum GPU inference time on a single NVIDIA RTX 6000 Ada GPU after a warmup period. All methods are compared at B=100. To trace MS-GPT across candidate-pool sizes from 100 to 1500, we keep the 20 query groups fixed and increase the number of samples per group. We measure MS-GPT, MS-BART, MIST+MolForge, and MADGEN directly; the FRIGID and DiffMS timings are taken from FRIGID [2], which reports measurements on the same GPU.

B.6 Comparability of Reported MCES Distances

The baseline Top-k MCES values in Table 1 are taken from the respective papers. Their stopping thresholds, molecular preprocessing, failure handling, and solver time limits differ, as summarized in Table 8. These choices affect whether MCES returns an exact

Table 8: MCES configurations in the released baseline evaluators. PuLP denotes the first available solver; time limits apply to each molecular pair, and none denotes no explicit time limit. Bond sum denotes $|E_1| + |E_2|$. †MADGEN assigns 100 to an invalid candidate and zero to an invalid target SMILES.

Method	Threshold	Molecular preprocessing	Fallback (Failed pair / No candidates)	Solver (time limit)
MADGEN [43]	20	raw SMILES	100 †	HiGHS (10 s)
MS-BART [14]	15	canonical SMILES	15 / 100	PuLP (none)
DiffMS [3]	100	tautomer-canonicalized SMILES	bond sum / 100	PuLP (none)
FRIGID [2]	15	stereo-free, tautomer-canonicalized SMILES	bond sum / 100	PuLP (600 s)

optimum or a lower bound, which molecular graphs are compared, and how failed comparisons are scored; the paper-reported values are therefore not strictly comparable.

MCES settings for MS-GPT. Our Table 1 entries follow the DiffMS configuration. For MCES, the complete candidate pool is grouped by InChIKey-14 and ranked by generation frequency. After tautomer canonicalization and stereochemistry removal, candidate–target pairs are evaluated with threshold 100 and stronger-bound selection disabled, using the first available PuLP solver without an explicit time limit. Failed pairs receive the bond-count sum $|E_{\text{true}}| + |E_{\text{pred}}|$, and spectra without candidates receive 100. MCES@k is the minimum over the top-k candidates, averaged over each full test set.

C Case Studies

Under the fixed-grid inference and ranking protocol in Appendix B.5, Figures 13 and 14 examine representative successes and failures as candidate-pool size B increases. They show when candidate-pool scaling changes Top-1 and distinguish finite-sample competition from frequency-dominated misranking, missing candidate support, posterior mismatch, and fingerprint resolution limits.

C.1 Candidate-Pool Scaling and Candidate Recovery

The stable rows in Figures 13(a) and 14(a) show that imperfect posterior evidence need not preclude an exact Top-1 prediction. With *Fidelity* values of 0.909 and 0.800, respectively, both targets are Top-1 at B=100 and remain so through B=1500. More revealingly, the stable and recovery rows on NPLIB1 share the same *Fidelity*, yet only the latter changes leader; fidelity alone therefore does not determine sensitivity to candidate-pool size.

In the recovery rows, the decoys lead by only 8:5 and 41:36 at B=100, while the targets have higher posterior similarity. Both targets become Top-1 at B=200 and remain there at B=1500. These trajectories are consistent with finite-sample near-ties: additional draws can overturn a narrow frequency lead. The cases therefore complement the aggregate result in Appendix A.4: although larger pools primarily improve candidate-pool recall, they can also change Top-1 when the target is already present and the leading generation counts are close.

C.2 Decoy Preference and Candidate Support

The two NPLIB1 failures separate misranking from failure to generate the target. In Figure 13(c), the target is absent at B=100 and reaches rank 2 by B=200, but Top-1 never changes. Although the target is distinguishable from the decoy (*Tani.* = 0.83) and better aligned with the posterior (0.74 versus 0.67), the decoy leads 469:70 at B=1500. Thus, a larger B improves candidate-pool recall, while the decoy's persistent frequency advantage preserves the wrong leader.

In Figure 13(d), the target remains absent through B=1500. The leading prediction replaces the seven-membered ring with a six-membered ring plus a separate [CH_2], satisfying the elemental composition but exposing the lack of a hard connectivity constraint during token generation.

C.3 Posterior Quality and Fingerprint Resolution

Figure 14(d) exposes a low-fidelity posterior mismatch. *Fidelity* is 0.070, yet a nearly unrelated prediction (*Tani.* = 0.06) reaches *Post.* = 0.87, while the target is never generated. The posterior is thus "confidently wrong" only operationally: the query band matches the wrong fingerprint neighborhood, and increasing B reinforces the resulting prediction but cannot alter the underlying posterior mismatch.

Figure 14(c) isolates a different failure: the query band reaches the oracle radius-2 fingerprint, but that representation itself is non-identifying. *Fidelity*, *Post.*, and *Tani.* are all 1.00, yet the target remains second at every candidate-pool size. The target and hydroxyl-relocation decoy have identical radius-2 Morgan fingerprints but become distinguishable at radius 3. Table 9 contrasts this pair with the NPLIB1 case in Figure 13(c), which is already distinguishable at radius 2.

This non-identifiability persists even under the oracle radius-2 fingerprint, making the failure orthogonal to the oracle-to-posterior gap and providing a concrete representational limit of the fixed-dimensional fingerprint posterior used by MS-GPT. A higher-radius or multiscale condition can encode this distinction, but whether the additional detail can be inferred reliably from spectra remains open.

Table 9: Target–decoy Morgan Tanimoto similarity across fingerprint radii for Figures 14(c) and 13(c). The MassSpecGym pair is fingerprint-equivalent at radius 2 but becomes distinguishable at larger radii; the NPLIB1 pair is already distinguishable at radius 2.

Spectrum	Structural change	r=2	r=3	r=4
MassSpecGymID 0221321	Hydroxyl relocation	1.00	0.95	0.86
CCMSLIB 00000848359	Side-chain relocation	0.83	0.64	0.54

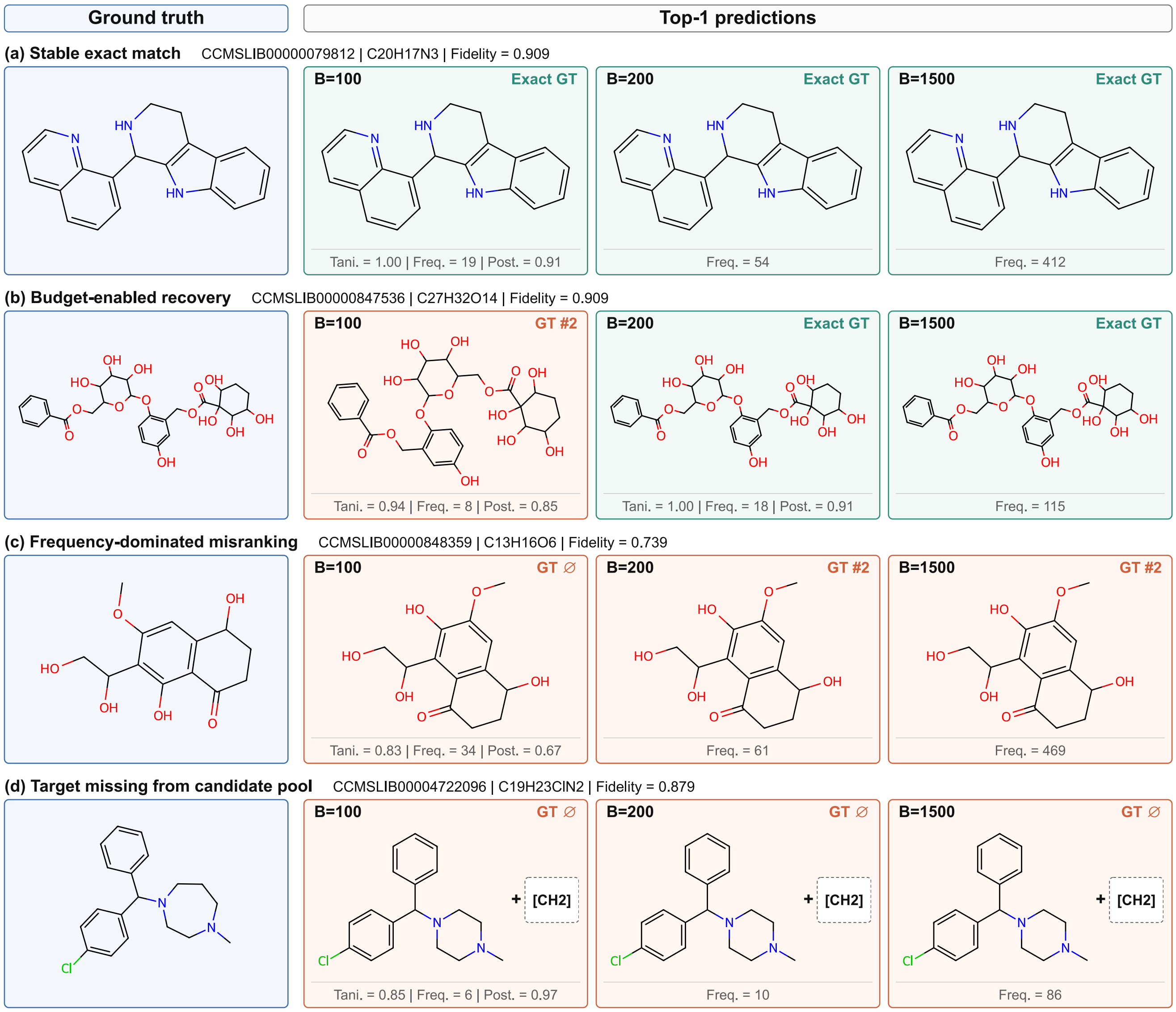

Figure 13: MS-GPT case studies on NPLIB1 across candidate-pool sizes B. Each row compares the ground truth (blue) with Top-1 predictions (green: exact; orange: decoy). *Fidelity* and *Post.* are, respectively, the maximum target–query and prediction–query radius-2 Morgan similarities over the active-bit density band; *Tani.* is the prediction–target radius-2 Morgan similarity, and *Freq.* is the grouped generation count. For a repeated Top-1 structure, only *Freq.* is repeated because the other scores are unchanged. "Exact GT," "GT #k," and "GT ∅" denote an exact leading match, target rank, and target absence. The dashed box in (d) contains the separate $[CH_2]$ component.

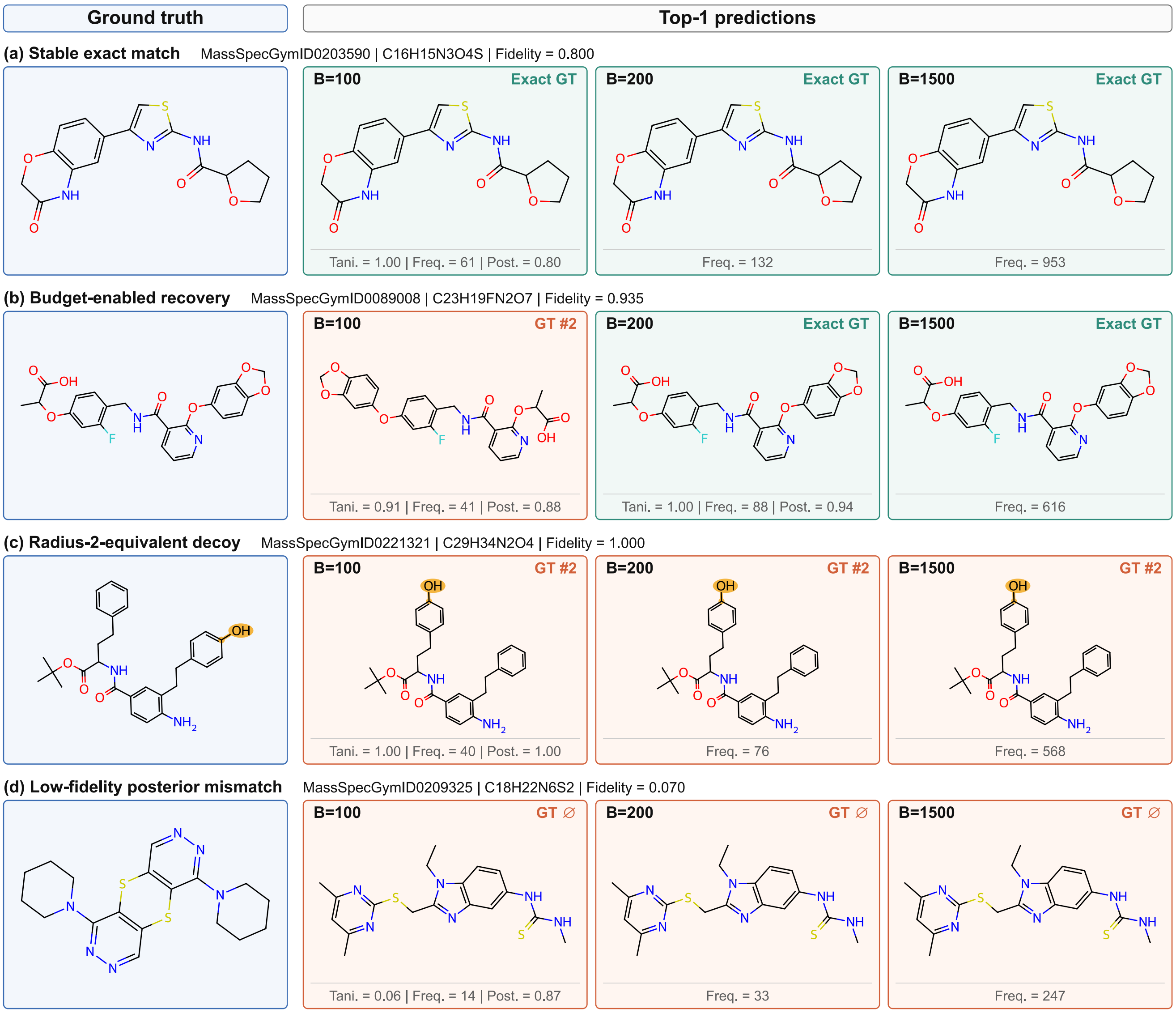

Figure 14: MS-GPT case studies on MassSpecGym across candidate-pool sizes B. Each row compares the ground truth (blue) with Top-1 predictions (green: exact; orange: decoy). *Fidelity* and *Post.* are, respectively, the maximum target–query and prediction–query radius-2 Morgan similarities over the active-bit density band; *Tani.* is the prediction–target radius-2 Morgan similarity, and *Freq.* is the grouped generation count. For a repeated Top-1 structure, only *Freq.* is repeated because the other scores are unchanged. "Exact GT," "GT #k," and "GT ∅" denote an exact leading match, target rank, and target absence. The gold highlight marks the migrated hydroxyl group.